Lesson 6: Conditional Expectation and the Tower Property

AI/ML Mathematics

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1 Motivation and Overview

The value function in reinforcement learning is defined as $V^\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s]$, a conditional expectation. Every Bellman equation—the recursive backbone of dynamic programming and temporal-difference learning—is derived by splitting a conditional expectation across two successive time steps. The algebraic engine behind this splitting is the *tower property* (law of total expectation):

$$\mathbb{E}[X] = \mathbb{E}\big[\mathbb{E}[X \mid Y]\big].$$

This lesson develops conditional expectation rigorously, establishes its main algebraic properties (linearity, pull-out, monotonicity, tower), and shows how the tower property, combined with the Markov property of MDPs, produces the Bellman expectation equation.

Why This Matters for RL

The tower property is the single most-used probabilistic tool in RL theory. It enables:

- The derivation of the Bellman equation from the return definition.
- The proof that the TD error δ_t^π has zero conditional mean (a martingale difference property).
- The expansion of multi-step expectations into single-step sums over actions and next states.

Prerequisites. Lesson 4 (probability spaces) and Lesson 5 (random variables, expectation, linearity of expectation, conditional probability). We assume throughout that $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space and all random variables are discrete unless otherwise stated.

2 Conditional Expectation Given an Event

Definition 2.1 (Conditional Expectation Given an Event)**.** Let X be a random variable with $\mathbb{E}[|X|] < \infty$ and let $B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$. The *conditional expectation of X given B* is the number

$$\mathbb{E}[X \mid B] = \frac{\mathbb{E}[X \cdot \mathbf{1}_B]}{\mathbb{P}(B)} = \sum_x x\,\mathbb{P}(X = x \mid B) = \sum_x x\,\frac{\mathbb{P}(X = x,\, B)}{\mathbb{P}(B)}. \tag{1}$$

Remark 2.2. $\mathbb{E}[X \mid B]$ is a *number*, not a random variable. It answers the question: "What is the average value of X among outcomes in B?"

Example 2.3. Let X be the outcome of a fair six-sided die and $B = \{X \text{ is even}\} = \{2, 4, 6\}$. Then $\mathbb{P}(B) = 1/2$ and

$$\mathbb{E}[X \mid B] = \frac{2 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6}}{1/2} = \frac{12/6}{1/2} = 4.$$

Proposition 2.4 (Conditional Expectation via Total Probability)**.** *If $B_1, B_2, \ldots, B_n$ partition Ω with $\mathbb{P}(B_i) > 0$, then*

$$\mathbb{E}[X] = \sum_{i=1}^n \mathbb{E}[X \mid B_i]\,\mathbb{P}(B_i). \tag{2}$$

Proof. Expanding each conditional expectation:

$$\begin{aligned}\sum_{i=1}^n \mathbb{E}[X \mid B_i]\,\mathbb{P}(B_i) &= \sum_{i=1}^n \left(\sum_x x\,\frac{\mathbb{P}(X = x, B_i)}{\mathbb{P}(B_i)}\right)\mathbb{P}(B_i) \\ &= \sum_{i=1}^n \sum_x x\,\mathbb{P}(X = x, B_i) = \sum_x x \sum_{i=1}^n \mathbb{P}(X = x, B_i). \end{aligned} \tag{3}$$

Since $\{B_i\}$ partitions Ω, $\sum_i \mathbb{P}(X = x, B_i) = \mathbb{P}(X = x)$, giving $\sum_x x\,\mathbb{P}(X = x) = \mathbb{E}[X]$. □

3 Conditional Expectation Given a Random Variable

Definition 3.1 (Conditional Expectation Given a Random Variable)**.** For discrete random variables X and Y with $\mathbb{E}[|X|] < \infty$, the *conditional expectation of X given Y* is the random variable $\mathbb{E}[X \mid Y]$ defined by

$$\mathbb{E}[X \mid Y](\omega) = \mathbb{E}[X \mid Y = Y(\omega)] = \sum_x x\,\mathbb{P}(X = x \mid Y = Y(\omega)). \tag{4}$$

Equivalently, $\mathbb{E}[X \mid Y] = g(Y)$ where $g(y) = \mathbb{E}[X \mid Y = y]$ for each y in the range of Y.

Remark 3.2 (Key conceptual point). Unlike $\mathbb{E}[X \mid B]$ (a number), $\mathbb{E}[X \mid Y]$ is a *random variable*: its value depends on the outcome ω through $Y(\omega)$. It is always a function of Y, hence $\sigma(Y)$-measurable.

Definition 3.3 (Conditional Expectation Given a σ-Algebra)**.** More generally, if $\mathcal{G} \subseteq \mathcal{F}$ is a sub-σ-algebra, the *conditional expectation* $\mathbb{E}[X \mid \mathcal{G}]$ is the unique (a.s.) $\mathcal{G}$-measurable random variable satisfying

$$\int_G \mathbb{E}[X \mid \mathcal{G}]\, d\mathbb{P} = \int_G X\, d\mathbb{P} \quad \text{for all } G \in \mathcal{G}. \tag{5}$$

Existence and uniqueness (up to a.s. equality) follow from the Radon–Nikodym theorem [Durrett, 2019, Billingsley, 1995]. When $\mathcal{G} = \sigma(Y)$, this recovers Definition 3.1.

Connection to RL

The value function $V^\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s]$ is $\mathbb{E}_\pi[G_t \mid S_t]$ evaluated at $S_t = s$. Equivalently, $V^\pi(S_t) = \mathbb{E}_\pi[G_t \mid S_t]$ is the random variable that assigns to each trajectory the expected return from the state visited at time t. The fact that $V^\pi(S_t)$ is $\sigma(S_t)$-measurable is precisely the statement that it is a function of S_t alone.

4 Properties of Conditional Expectation

Throughout this section, X, Y, Z are discrete random variables with $\mathbb{E}[|X|], \mathbb{E}[|Y|] < \infty$.

4.1 Linearity

Theorem 4.1 (Linearity of Conditional Expectation)**.** *For constants $a, b \in \mathbb{R}$:*

$$\mathbb{E}[aX + bY \mid Z] = a\,\mathbb{E}[X \mid Z] + b\,\mathbb{E}[Y \mid Z]. \tag{6}$$

Proof. Fix $Z = z$ with $\mathbb{P}(Z = z) > 0$. Expanding by definition:

$$\begin{aligned}\mathbb{E}[aX + bY \mid Z = z] &= \sum_{x,y} (ax + by)\,\mathbb{P}(X = x, Y = y \mid Z = z) \\ &= a \sum_{x,y} x\,\mathbb{P}(X = x, Y = y \mid Z = z) + b \sum_{x,y} y\,\mathbb{P}(X = x, Y = y \mid Z = z).\end{aligned} \tag{7}$$

We marginalize the first sum over y:

$$\sum_{x,y} x\,\mathbb{P}(X = x, Y = y \mid Z = z) = \sum_x x \sum_y \mathbb{P}(X = x, Y = y \mid Z = z) = \sum_x x\,\mathbb{P}(X = x \mid Z = z) = \mathbb{E}[X \mid Z = z]. \tag{8}$$

Similarly, the second sum equals $\mathbb{E}[Y \mid Z = z]$. Since the identity holds for every z, we have $\mathbb{E}[aX + bY \mid Z] = a\,\mathbb{E}[X \mid Z] + b\,\mathbb{E}[Y \mid Z]$ as random variables. □

4.2 Pull-Out Property

Pull-Out Property

Theorem 4.2 (Pull-Out Property)**.** *If $h(Z)$ is a function of Z (i.e., $\sigma(Z)$-measurable), then*

$$\mathbb{E}[h(Z) \cdot X \mid Z] = h(Z) \cdot \mathbb{E}[X \mid Z]. \tag{9}$$

Proof. Fix $Z = z$. Then $h(Z) = h(z)$ is a constant under this conditioning:

$$\begin{aligned}\mathbb{E}[h(Z) \cdot X \mid Z = z] &= \sum_x h(z) \cdot x \cdot \mathbb{P}(X = x \mid Z = z) \\ &= h(z) \sum_x x\, \mathbb{P}(X = x \mid Z = z) = h(z) \cdot \mathbb{E}[X \mid Z = z].\end{aligned} \tag{10}$$

Since this holds for every z, the identity holds as random variables. □

Connection to RL

When proving that the TD error has zero conditional mean, we condition on $S_t = s$ and pull out $V(S_t) = V(s)$:

$$\mathbb{E}[\delta_t \mid S_t = s] = \mathbb{E}[R_{t+1} + \gamma V(S_{t+1}) \mid S_t = s] - V(s).$$

The term $V(s)$ exits the expectation because it is fully determined by S_t. This is the pull-out property with $h(S_t) = V(S_t)$ and $Z = S_t$.

4.3 Monotonicity

Proposition 4.3 (Monotonicity of Conditional Expectation)**.** *If $X \leq Y$ almost surely, then $\mathbb{E}[X \mid Z] \leq \mathbb{E}[Y \mid Z]$ almost surely.*

Proof. Fix $Z = z$. Since $X \leq Y$ a.s., we have $\mathbb{P}(X \leq Y \mid Z = z) = 1$ (because $\mathbb{P}(X > Y) = 0$ implies $\mathbb{P}(X > Y, Z = z) = 0$). Therefore:

$$\mathbb{E}[Y \mid Z = z] - \mathbb{E}[X \mid Z = z] = \mathbb{E}[Y - X \mid Z = z] \geq 0, \tag{11}$$

where the equality uses linearity and the inequality uses that $Y - X \geq 0$ a.s. conditionally. □

4.4 Conditional Expectation of Measurable Functions

Proposition 4.4 (Conditioning on Known Information)**.** *If X is $\sigma(Z)$-measurable (i.e., $X = g(Z)$ for some function g), then $\mathbb{E}[X \mid Z] = X$ almost surely.*

Proof. If $X = g(Z)$, then fixing $Z = z$ gives $X = g(z)$, which is constant. Hence $\mathbb{E}[X \mid Z = z] = g(z) = X$ on the event $\{Z = z\}$. □

Corollary 4.5. *If c is a constant, $\mathbb{E}[c \mid Z] = c$.*

5 The Tower Property (Law of Total Expectation)

Law of Total Expectation (Tower Property)

Theorem 5.1 (Law of Total Expectation). *For discrete random variables X and Y with $\mathbb{E}[|X|] < \infty$:*

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]] = \sum_y \mathbb{E}[X \mid Y = y]\,\mathbb{P}(Y = y). \tag{12}$$

Proof. We expand the right-hand side and show it equals $\mathbb{E}[X]$.

Step 1: Expand the inner conditional expectation.

$$\sum_y \mathbb{E}[X \mid Y = y]\,\mathbb{P}(Y = y) = \sum_y \left(\sum_x x\,\mathbb{P}(X = x \mid Y = y) \right) \mathbb{P}(Y = y). \tag{13}$$

Step 2: Use the definition of conditional probability.

$$= \sum_y \sum_x x \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)} \cdot \mathbb{P}(Y = y) = \sum_y \sum_x x\,\mathbb{P}(X = x, Y = y). \tag{14}$$

Step 3: Interchange sums and marginalize.

$$= \sum_x x \sum_y \mathbb{P}(X = x, Y = y) = \sum_x x\,\mathbb{P}(X = x) = \mathbb{E}[X]. \tag{15}$$

The interchange of summation is justified by absolute convergence (since $\mathbb{E}[|X|] < \infty$), and the marginalization uses $\sum_y \mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x)$. □

Remark 5.2 (Interpretation). The tower property says: to compute $\mathbb{E}[X]$, we may first compute $\mathbb{E}[X \mid Y]$ (which is a random variable, a function of Y) and then average over Y. The "tower" name comes from thinking of $\mathbb{E}\big[\mathbb{E}[\cdot \mid \cdot]\big]$ as a stack of two operations.

Theorem 5.3 (Generalized Tower Property). *If $\mathcal{G} \subseteq \mathcal{H} \subseteq \mathcal{F}$ are sub-σ-algebras, then*

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{H}] \mid \mathcal{G}] = \mathbb{E}[X \mid \mathcal{G}]. \tag{16}$$

In particular, for random variables with $\sigma(Z) \subseteq \sigma(Y, Z)$:

$$\mathbb{E}[\mathbb{E}[X \mid Y, Z] \mid Z] = \mathbb{E}[X \mid Z]. \tag{17}$$

Proof. We prove (17) in the discrete case. Fix $Z = z$ with $\mathbb{P}(Z = z) > 0$:

$$\begin{aligned}
\mathbb{E}[\mathbb{E}[X \mid Y, Z] \mid Z = z] &= \sum_y \mathbb{E}[X \mid Y = y, Z = z]\,\mathbb{P}(Y = y \mid Z = z) \\
&= \sum_y \left(\sum_x x\,\mathbb{P}(X = x \mid Y = y, Z = z) \right) \mathbb{P}(Y = y \mid Z = z) \\
&= \sum_x x \sum_y \mathbb{P}(X = x \mid Y = y, Z = z)\,\mathbb{P}(Y = y \mid Z = z).
\end{aligned} \tag{18}$$

Applying the chain rule to the inner product:

$$\begin{aligned}
\mathbb{P}(X = x \mid Y = y, Z = z)\,\mathbb{P}(Y = y \mid Z = z) &= \frac{\mathbb{P}(X = x, Y = y, Z = z)}{\mathbb{P}(Y = y, Z = z)} \cdot \frac{\mathbb{P}(Y = y, Z = z)}{\mathbb{P}(Z = z)} \\
&= \frac{\mathbb{P}(X = x, Y = y, Z = z)}{\mathbb{P}(Z = z)}.
\end{aligned} \tag{19}$$

Summing over y:

$$\sum_y \frac{\mathbb{P}(X = x, Y = y, Z = z)}{\mathbb{P}(Z = z)} = \frac{\mathbb{P}(X = x, Z = z)}{\mathbb{P}(Z = z)} = \mathbb{P}(X = x \mid Z = z). \tag{20}$$

Substituting back: $\mathbb{E}[\mathbb{E}[X \mid Y, Z] \mid Z = z] = \sum_x x\,\mathbb{P}(X = x \mid Z = z) = \mathbb{E}[X \mid Z = z]$. □

Remark 5.4. The generalized tower property says: if we first condition on "more information" ($\mathcal{H}$) and then average back to "less information" ($\mathcal{G} \subseteq \mathcal{H}$), the result is the same as conditioning on the less informative σ-algebra directly. Conditioning on more information and then averaging away the excess is the same as never having conditioned on the excess.

6 MDP Application: Deriving the Bellman Equation

We now apply the tower property and MDP expansion to derive the Bellman expectation equation, the central recursive equation of reinforcement learning.

6.1 MDP Expectation Expansion

Theorem 6.1 (MDP Expectation Expansion)**.** *Let $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$ be a finite MDP with policy π. For any function $f : \mathcal{A} \times \mathcal{S} \times \mathbb{R} \to \mathbb{R}$:*

$$\mathbb{E}_\pi[f(A_t, S_{t+1}, R_{t+1}) \mid S_t = s] = \sum_{a \in \mathcal{A}} \pi(a \mid s) \sum_{s' \in \mathcal{S}} P(s' \mid s, a)\, f(a, s', r(s, a, s')). \tag{21}$$

Proof. By the definition of conditional expectation:

$$\begin{aligned} &\mathbb{E}_\pi[f(A_t, S_{t+1}, R_{t+1}) \mid S_t = s] \\ &\quad = \sum_{a \in \mathcal{A}} \sum_{s' \in \mathcal{S}} f(a, s', r(s, a, s'))\, \mathbb{P}_\pi(A_t = a, S_{t+1} = s' \mid S_t = s). \end{aligned} \tag{22}$$

By the chain rule for conditional probability and the MDP dynamics:

$$\begin{aligned} \mathbb{P}_\pi(A_t = a, S_{t+1} = s' \mid S_t = s) &= \mathbb{P}_\pi(A_t = a \mid S_t = s) \cdot \mathbb{P}(S_{t+1} = s' \mid S_t = s, A_t = a) \\ &= \pi(a \mid s) \cdot P(s' \mid s, a). \end{aligned} \tag{23}$$

Substituting (23) into (22) gives (21). □

6.2 The Bellman Expectation Equation

Bellman Expectation Equation

Corollary 6.2 (Bellman Expectation Equation)**.** *For the value function $V^\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s]$ with $G_t = R_{t+1} + \gamma G_{t+1}$:*

$$V^\pi(s) = \sum_a \pi(a \mid s) \sum_{s'} P(s' \mid s, a) \left[r(s, a, s') + \gamma V^\pi(s')\right]. \tag{24}$$

Proof. **Step 1: Decompose the return.** By the definition of the discounted return:

$$V^\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s] = \mathbb{E}_\pi[R_{t+1} + \gamma G_{t+1} \mid S_t = s]. \tag{25}$$

Step 2: Apply linearity of conditional expectation.

$$= \mathbb{E}_\pi[R_{t+1} \mid S_t = s] + \gamma\, \mathbb{E}_\pi[G_{t+1} \mid S_t = s]. \tag{26}$$

Step 3: Apply the tower property to the second term. By the generalized tower property (Theorem 5.3) and the Markov property:

$$\begin{aligned}\mathbb{E}_\pi[G_{t+1} \mid S_t = s] &= \mathbb{E}_\pi[\mathbb{E}_\pi[G_{t+1} \mid S_{t+1}] \mid S_t = s] \\ &= \mathbb{E}_\pi[V^\pi(S_{t+1}) \mid S_t = s]. \end{aligned} \tag{27}$$

The inner conditioning uses the Markov property: given S_{t+1}, G_{t+1} is conditionally independent of S_t, so $\mathbb{E}_\pi[G_{t+1} \mid S_{t+1}, S_t] = \mathbb{E}_\pi[G_{t+1} \mid S_{t+1}] = V^\pi(S_{t+1})$.

Step 4: Expand via MDP dynamics. Combining (26) and (27):

$$V^\pi(s) = \mathbb{E}_\pi[R_{t+1} + \gamma V^\pi(S_{t+1}) \mid S_t = s]. \tag{28}$$

Setting $f(a, s', r) = r + \gamma V^\pi(s')$ in Theorem 6.1:

$$V^\pi(s) = \sum_a \pi(a \mid s) \sum_{s'} P(s' \mid s, a) \left[r(s, a, s') + \gamma V^\pi(s')\right]. \tag{29}$$

□

Remark 6.3 (Operator notation). In operator notation, the Bellman equation reads $V^\pi = \mathcal{T}^\pi V^\pi$, where the Bellman operator $\mathcal{T}^\pi$ is defined by the right-hand side of (24). The fact that V^π is a fixed point of $\mathcal{T}^\pi$ is a direct consequence of the tower property.

Connection to TD Learning

The Bellman equation derivation relies on three tools from this lesson:

1. **Linearity** (Theorem 4.1) to split $\mathbb{E}[R_{t+1} + \gamma G_{t+1}]$.
2. **Tower property** (Theorem 5.3) to replace $\mathbb{E}[G_{t+1} \mid S_t]$ with $\mathbb{E}[V^\pi(S_{t+1}) \mid S_t]$.
3. **MDP expansion** (Theorem 6.1) to write the result as an explicit sum over actions and next states.

TD learning bootstraps by replacing the unknown $V^\pi(S_{t+1})$ with a current estimate $V(S_{t+1})$, producing the TD target $R_{t+1} + \gamma V(S_{t+1})$. The tower property guarantees that $\mathbb{E}_\pi[\delta_t^\pi \mid S_t = s] = 0$ when $V = V^\pi$, the foundation of all TD convergence proofs.

7 Exercises

Exercise 7.1. Let X and Y be independent with $X \sim \text{Bernoulli}(p)$ and $Y \sim \text{Bernoulli}(q)$. Define $Z = X + Y$. Compute $\mathbb{E}[X \mid Z]$ for each possible value of Z.

Exercise 7.2. Let X and Y be random variables with finite second moments. Using the pull-out property and the tower property, show that $\mathbb{E}[Y \cdot \mathbb{E}[X \mid Y]] = \mathbb{E}[XY]$. *Hint:* Start from the right-hand side and apply the tower property with the inner conditioning on Y.

Exercise 7.3. Let X, Y, Z be discrete random variables with $\mathbb{E}[|X|] < \infty$. Show that $\mathbb{E}\big[\mathbb{E}[\mathbb{E}[X \mid Y, Z] \mid Y]\big] = \mathbb{E}[X]$ by applying the tower property twice.

Exercise 7.4. Consider a two-state MDP with $\mathcal{S} = \{s_1, s_2\}$, $\mathcal{A} = \{a\}$ (single action), $P(s_1 \mid s_1, a) = 0.6$, $P(s_2 \mid s_1, a) = 0.4$, $P(s_1 \mid s_2, a) = 0.3$, $P(s_2 \mid s_2, a) = 0.7$, $r(s_1, a, \cdot) = 1$, $r(s_2, a, \cdot) = 2$, and $\gamma = 0.9$. Write the Bellman equation (24) for each state and solve the resulting 2×2 linear system for $V^\pi(s_1)$ and $V^\pi(s_2)$.

Exercise 7.5. Prove the *conditional tower property*: if $\mathcal{G} \subseteq \mathcal{H}$ are sub-σ-algebras, then for any event $A \in \mathcal{G}$:

$$\mathbb{E}\big[\mathbb{E}[X \mid \mathcal{H}] \cdot \mathbf{1}_A\big] = \mathbb{E}[X \cdot \mathbf{1}_A].$$

Hint: Use the defining property (5) of conditional expectation.

Exercise 7.6. Let X and Y be independent with $\mathbb{E}[|X|] < \infty$. Show that $\mathbb{E}[X \mid Y] = \mathbb{E}[X]$ a.s. *Hint:* For independent discrete RVs, $\mathbb{P}(X = x \mid Y = y) = \mathbb{P}(X = x)$.

Exercise 7.7. In an MDP with policy π and true value function V^π, the TD error is $\delta_t^\pi = R_{t+1} + \gamma V^\pi(S_{t+1}) - V^\pi(S_t)$. Using linearity, the pull-out property, and the Bellman equation, show that $\mathbb{E}_\pi[\delta_t^\pi \mid S_t = s] = 0$.

8 Summary

Result	**Statement**	**RL Role**
Cond. exp. given event	$\mathbb{E}[X \mid B] = \mathbb{E}[X\mathbf{1}_B]/\mathbb{P}(B)$	State-conditioned values
Cond. exp. given RV	$\mathbb{E}[X \mid Y] = g(Y)$, a random variable	$V^\pi(S_t) = \mathbb{E}[G_t \mid S_t]$
Linearity	$\mathbb{E}[aX + bY \mid Z] = a\mathbb{E}[X \mid Z] + b\mathbb{E}[Y \mid Z]$	Splitting $G_t = R + \gamma G$
Pull-out property	$\mathbb{E}[h(Z)X \mid Z] = h(Z)\mathbb{E}[X \mid Z]$	Extracting $V(S_t)$
Tower property	$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]]$	Bellman recursion
Generalized tower	$\mathbb{E}[\mathbb{E}[X \mid \mathcal{H}] \mid \mathcal{G}] = \mathbb{E}[X \mid \mathcal{G}]$	Multi-step bootstrapping
MDP expansion	$\mathbb{E}_\pi[f(A, S', R') \mid S = s] = \sum_a \pi \sum_{s'} P f$	Explicit Bellman sums
Bellman equation	$V^\pi(s) = \sum_a \pi \sum_{s'} P[r + \gamma V^\pi]$	Fixed-point of $\mathcal{T}^\pi$

Next lesson: Variance, Covariance, and Independence — the tools for analyzing the spread and correlation of random variables, culminating in the law of total variance that explains why TD learning has lower variance than Monte Carlo.

References

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